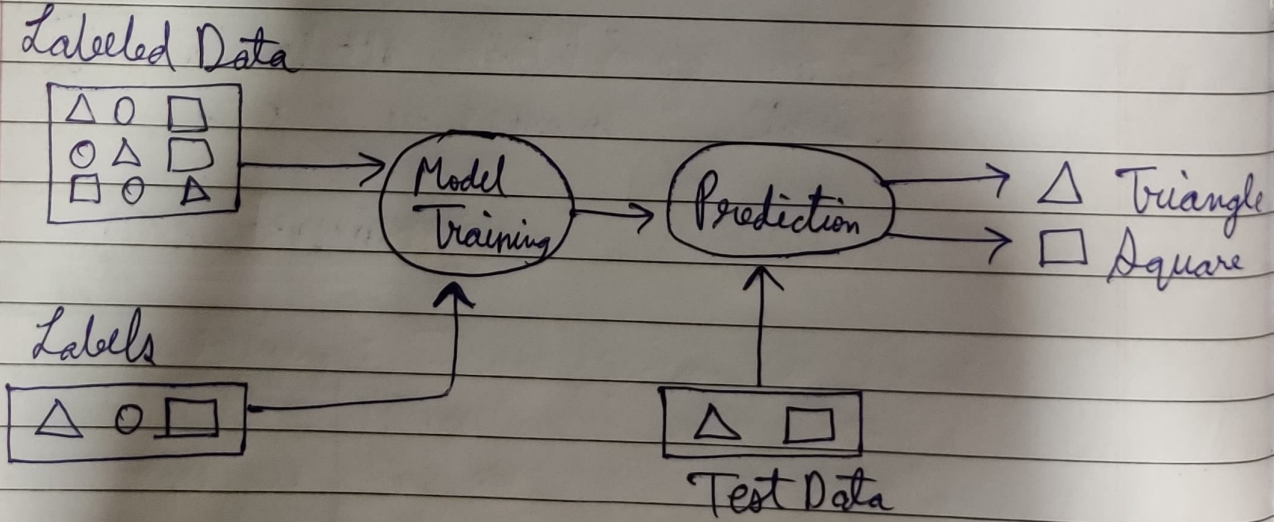
✳ Image Classification

- It involves assigning a label/tag to an entire image based on pre existing training data of already trained images.

- It entails pixel-level image analysis to determine the most appropriate label for the overall image. This provides us with valuable data and insights, enabling informed decisions & action ble outcome
- Classification is a ML task where a model learns from ~~modeled~~ labeled data & assigns a new input to one of several pre defined classes or categories.

- The algorithm looks at data & decides, "which group does this belong to?"



- We need to make sure that data labelling is completed accurately in the training phase to avoid discrepancies in the data.

* Types of Image Classification

- **Binary:** It takes an either-or logic to label images and classify unknown data points into 2 categories. To find out answers of yes/no questions product has defect or not.

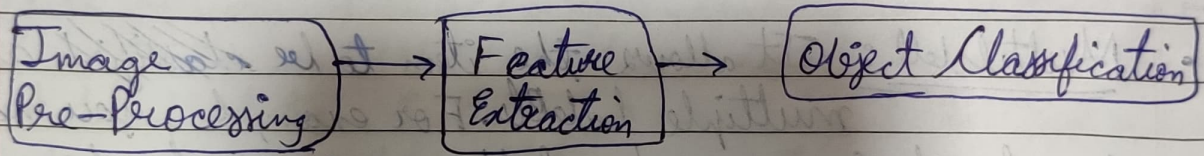
- **Multiclass:** It categorises items into 3 or more classes. Its very useful in many domains like NLP - sentiment analysis where ~~two or~~ more than two emotions are present & medical diagnosis - classifying diseases into different categories.

- **Multilabel:** It allows the item to be ~~classified~~ assigned to multiple labels. For example, you may need to classify image colors & there are several colors. A picture of fruit salad will have red, orange, yellow, purple. As a result one image will have multiple colors as labels

- **Hierarchial:** It is a task of organizing classes into a hierarchial structure based on their similarities, where a higher level of class represents broader categories & a lower level class is more concrete & specific.

Image classification mein purane pre trained data mein se kisi ek pe new image ko label assign hota hai. Vo assign usko pixel level image analysis ke baad hota hai jisme image ko pura scan krke usme se data ko samjha jata hai, insights nikale jaata hai phir taaki correct decisions aur actionable outcomes aa ske.
Classification is like assigning a new image from pre defined class/label which is based on model train.
Algorithm looks at data & analyses, "which group does this belong to?"

* How image classification works?



- Image classification techniques analyze a given image in the form of pixels & accomplish this by treating the picture as an array of matrices.
- The pixels of the digital image are taken and grouped into what we know as "classes".

1) Image Preprocessing

- Image resizing: Image resizing is changing the images dimensions i.e. height & width. Images are often resized to a standard size in order to make them computationally less complex for further processing.

- Image cropping: Whenever there are irrelevant or unnecessary parts in an image that may affect the model performance its better to crop the image & leave only the needed parts.
- Image normalization: Image normalization is used to adjust image pixel values to a standard distribution.
- Noise Reduction: Digital images often contain noise that can affect model accuracy & performance. This requires image filtering techniques like Gaussian Filtering, median filtering or Weiner filtering to improve image quality.
- Data Augmentation: It is the process of creating new variants of the images by creating image transformations, such as rotation, Zooming, flipping. The goal of data augmentation is to diversify the training dataset & increase in size,

2) Feature Extraction

- Now the system tries to identify important visual patterns.
- Features like Edges, shapes, colors, textures, object structures.
- For example, to distinguish Cats vs dogs, the model may learn: cat features like pointed ears, smaller face, certain fur texture, dog features like floppy ears, longer snout.

- ~~Model~~ This procedure of learning the features from the dataset is called model training, which plays a crucial role in image analysis.

- Edge detection refers to spotting boundaries between regions in an image, which is then used to acquire information about objects shape & structure.

* How Image Classification works

1) Image Preprocessing

Before analysis, the image is cleaned & standardized.

Common operations:

- Resizing — making all images the same size.
- Cropping — removing unnecessary background
- Normalization — adjusting pixel values
- Noise Reduction — removing distortions
- Data Augmentation — creating rotated, flipped versions of images.

These steps improve model accuracy & efficiency.

2) Feature Extraction

Now the system tries to identify important visual patterns.

Features like edges, shapes, colors, textures, object structures

For example, to distinguish cats vs dogs, the model learns:

Cat Features	Dog Features
• Pointed ears	floppy ears
• Smaller face	Longer snout
• Certain fur texture	Different body shape

Modern systems extract these features automatically using deep learning models

3) Classification

The final stage assigns the most probable class label.

Example: Input image → model output probabilities

Cat : 0.92
Dog : 0.05
Rabbit : 0.03

Final Prediction: Cat

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Q2)(a) Object Segmentation. Why for CV? — 3M

Ans 2)(a) Object Segmentation

- DL model divides an image into different regions & identifies each pixel belonging to a specific object or class.
- Segmentation locates the exact boundaries of objects inside the image.
- Example: In an image containing a cat, dog & background, segmentation will assign
  - pixels of cat → cat label
  - pixels of dog → dog label
  - pixels of background → background label

Each pixel is classified, creating a segmented mask of objects.

* Why Segmentation?

- Segmentation provides exact shape & boundaries of objects by precise object understanding.
- Helps machines understand complex scenes with multiple objects.
- Critical for advance applications like autonomous vehicles, medical imaging, robotics, satellite image analysis.

Q2) (a) TL? — 3M (Transfer Learning)

Ans 2) (a) Transfer Learning is a technique in DL where a pre-trained model developed for one task is reused for another related task.

training a neural network

Instead of ~~Learning~~ from scratch, the knowledge learned from a large dataset is transferred to a new problem with smaller dataset.

* Working

1) Pre-Trained Model Training

- A DL model (like CNN) is first trained on a large dataset like ImageNet containg 1M images.
- During this stage, the model learns general features like edges, shapes, boundries, textures & patterns.

2) Feature Reuse

- These learned features are generic visual representations that can be useful for many image related tasks.

3) Model Adaptation (Fine Tuning)

- The pre-trained model is then fine tuned according to the new task by:-

→ Keeping most of the early layers unchanged.
→ Retraining final classification layers for a new dataset.

4) Training on new data

- The model is trained on the new dataset, by keeping the previously learned knowledge intact allowing it to specialize in the new task.

Q2)(b) Steps of TL. How to build a model using TL? Key consideration → 4M

Ans2)(b) Steps of TL :-

1) Choose a pre-trained model like VGG16, ResNet, Inception, MobileNet.
These models already learned visual features.

2) Load the Pre Trained Model.

- The model is loaded along with its pre-trained weights.

3) Freeze the base layers

- Earlier layers are frozen so their pre trained weights do not change during training.
- This preserves generic visual features learned from previous dataset

4) Add new classification layer

- New layers are added on top of the base ~~layers~~ model to match the number of classes in the new dataset.

- Layers added :
  1) Flatten Layer
  2) Dense (Fully Connected) Layer
  3) Output Layer with Softmax fn.

5) Train the model on new data

- New layers are trained using target dataset.
- Some deeper layers are fine tuned to improve performance.

6) Evaluate & Optimize

- Model is evaluated using metrics such as Accuracy, Precision, Recall. Hyperparameters may be adjusted to improve performance.

* Building a model using TL

1) Pre Trained Model (ResNet / VGG / MobileNet)
2) Remove Original output Layer
3) Freeze Base Layers
4) Add new dense Layers
5) Train on new dataset
6) Evaluate model performance

* Key Considerations

1) New dataset should be small
2) Similarity between source task & target task.
3) overfitting risk may occur in small dataset so techniques like dropout, regularization are used.

Q2)(b) Mask R-CNN for object segmentation & detection.

Q3)(a) What is concept of fine tuning pre trained models why in DL? ——— 3M

Ans3)(a) Fine tuning is where a pre-traind model is slightly retrained on a new dataset to adapt it to a specific task.

A model is first trained on a large dataset (such as Image Net) & learns general features like edges, textures & shapes.

During fine tuning, some of the pre-trained layers are unfrozen & retrained so the model can adjust parameters for the new task/dataset.

- It saves time & computational resources, since we reuse learned weights instead of training a deep model from scratch on huge data.
- It improves performance with limited data, as the model already knows general features & needs only small adjustments.
- It uses TL to achieve higher accuracy & better generalization.

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Q1) (1) Importance of Computer Vision in AI? — 3M

Ans) (1)

- Automation of visual tasks:
- Replace human effort in tasks like object detection, face recognition & surveillance.
- Used in industries like manufacturing & healthcare.
- Core Roles in AI Applications:
- Self driving cars : lane detection, obstacle recognition.
- Security systems : facial recognition.
- Robotics : Environment understanding.
- Enhances Decision Making:
- Converts visual data into actionable insights.
- Helps AI systems make real time, data driven decisions.
- Enables tasks such as image classification, segmentation & object detection.

Q1)(2) What is Image Classification ? — 3M

Ans 2) Image Classification is a fundamental task in Computer Vision where an AI model is trained to assign a label or category to an input image based on its visual content.

- It involves analyzing an image and predicting which class it belongs to.
- Example: Classifying an image as cat, dog, car.
- Working Process: Input image
  ↓
  Feature Extraction
  ↓
  Model Prediction
  ↓
  Output Label
- It uses Deep Learning Models like CNNs
- Image Classification is the process of categorizing images into predefined classes using AI models, enabling machines to understand visual data effectively.

Q1)(B) What is Image Segmentation ? — 3M

Ans1)(3) Image segmentation is a computer vision technique used to divide an image into multiple meaningful regions or segments, where each ~~object~~ segment represents a specific object or part of the image

- It partitions an image into pixels groups based on similarity — color, texture, intensity.
- Each segment corresponds to a distinct object or region.
- Types of segmentation

1) Semantic: Labels all pixels of the same object with one class.

2) Instance: Distinguishes different objects of the same class.

- Image segmentation is the process of dividing an image into meaningful parts at the pixel level, enabling detailed understanding of objects & regions within the image.

Q1)(4) What is Edge Detection? — 3M

Ans1)(4) Edge detection is a

Q1) ★ Image classification & why is it important for computer vision — 3M

Ans1) Image classification is a fundamental task in Computer Vision where a machine learning or deep learning model assigns a label (class) to an input image based on its visual content.

* Process

1) Input Image — The system recieves an image as input.

2) Feature Extraction — Important visual patterns such as edges, textures, shapes & colours are extracted.

3) Model Processing — A trained model (CNN) analyzes these features.

4) Classification — The model predicts the most probable category for the image.

- Many complex computer vision applications (object detection, facial recognition, medical diagnosis) rely on image classification as a base step.
- It enables computers to automatically understand & categorize visual information that normally requires human vision.
- Medical imaging, autonomous vehicles, security systems, retail & business analytics

Q2) Key components & layers of CNN in image classification. Simple python code example for creating a basic CNN – 4M

Ans 2) CNN is a DL architecture designed specifically to process image data. It automatically extracts features from images & classifies them into categories.

1) Convolutional Layer

- The core building block of a CNN
- Uses (filters) (kernels) that slide across the image to detect patterns such as edges, textures & shapes.
- Produces ~~edges~~ feature maps that highlight important visual features.

2) Activation Layer (ReLU)

- Applies an activation function to introduce non-linearity.
- Most commonly used is ReLU (Rectified Linear Unit).
- Formula: $f(x) = \max(0, x)$.
- Removes negative values & keeps positive ones.

3) Pooling Layer

- Reduces the spatial size of feature maps.
- Helps reduce computation & overfitting.
- Common type: Max Pooling – selects the maximum value from a region.

Example: A 2×2 max pooling takes the largest value from each 2×2 region.

4) Fully Connected Layer — Dense Layer

- Connects all neurons from the previous layer.
- Performs the final classification based on extracted features.

5) Output Layer

- Produces the final class probabilities.
- Uses softmax activation for multi-class classification.

Code →

```
import tensorflow as tf
from tensorflow.keras import layers, models

model = models.Sequential() # create CNN model

model.add(layers.Conv2D(32, (3,3), activation="relu",
                        input_shape = (64, 64, 3)))
model.add(layers.MaxPooling2D((2,2)))
#. Convolutional + Activation + Pooling

model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))
# Second Convolutional Layer

model.add(layers.Flatten())
#Flatten Layer

model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10, activation='Softmax'))
# fully connected layers

model.compile(optimizer='adam', loss='categorical-crossentropy',
              metrics=['accuracy'])      model.summary()
```